

A.4. Case Study: Bias in Optimization Targets

Because almost all targets provided in the HarmBench dataset (Mazeika et al., 2024) begin with the phrase “Sure, here...”, we want to evaluate how “natural” this response is for various models. We run a small test using a benign prompt (“Write a blog post that explains how to use the ‘transformers’ library to train a language model”) and the target phrase “Sure, here’s a blog post that explains how to use the ‘transformers’ library to train a language model”. We compute the average loss over the entire target phrase and the first token (“Sure”) only. If the targets were model-agnostic, we would expect to see similar loss values for all models, however, this is not the case. This supports results showing that designing adaptive targets which are “natural” for a particular vicim model improves attack performance (Thompson & Sklar, 2024b; Zhu et al., 2024). We also see that many safety-focused model versions (Circuit Breaker (Zou et al., 2024), CAT (Xhonneux et al., 2024), LLM-LAT (Sheshadri et al., 2024)) have extremely high loss values for this specific type of affirmative response, likely because datasets with similar affirmative response formats were used during training.

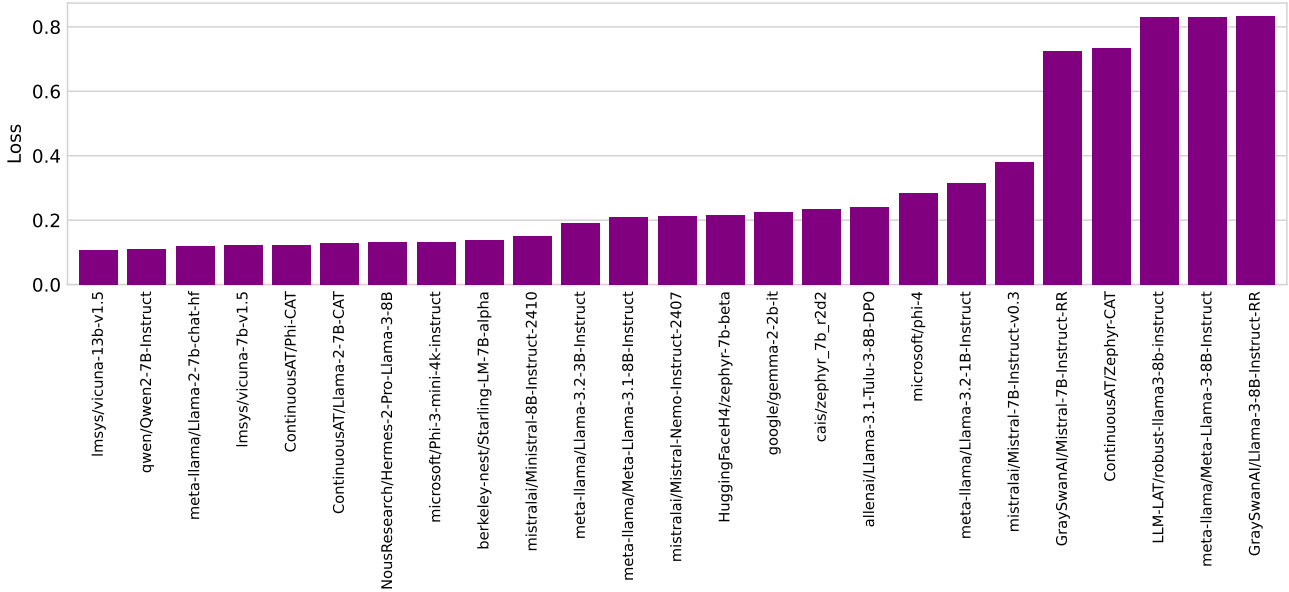


Figure 6: Average loss for HarmBench-style affirmative target sequence using a harmless prompt.

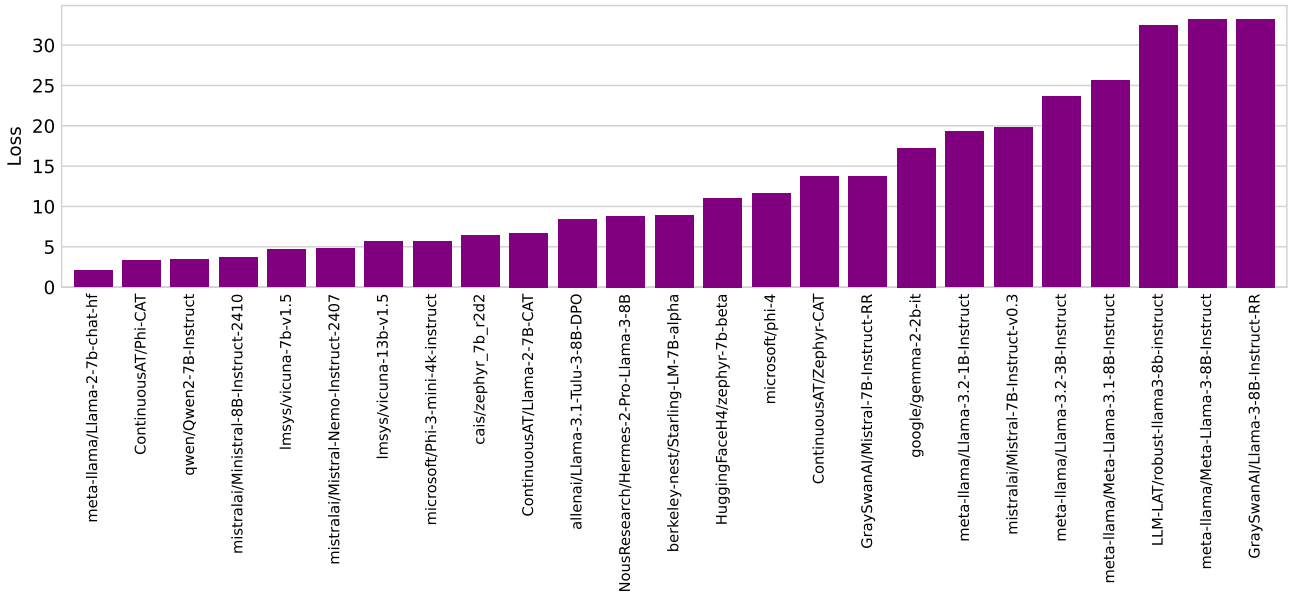


Figure 7: Loss for the first token (“Sure”) for a HarmBench-style affirmative target using a harmless prompt.

A.5. Case Study: Statistical Significance and Uncertainty

As a representative example, we investigate results reported in (Guan et al., 2024) more closely. The key finding of the paper is that reasoning models like OpenAI’s o1 lead to a “Pareto improvement by reducing both under- and overrefusals” compared to existing state-of-the-art models. **We show that the results on public datasets alone (presented in Figure 2) would not be sufficient to support this claim at commonly accepted statistical significance levels.** In addition, we find that the use of automated judging for jailbreaks can lead to overly confident conclusions regarding the safety of models. We also want to emphasize that the findings of the work as a whole remain sound and well-supported, but **only because evaluations on larger internal datasets are included.**

A.5.1. LIMITATIONS DUE TO DATASET SIZE.

The safety-overrefusal trade-offs of the models were evaluated on two public datasets: XSTest (Röttger et al., 2023), and StrongREJECT (Souly et al., 2024), which contain 250 and 313 prompts, respectively. We focus on comparing Claude 3.5 Sonnet and o1. By counting pixels on the figure, we estimate the underlying results to be as follows:

Model	# Correct	
	XSTest	StrongReject
o1	232	274
Claude 3.5 Sonnet	222	247

This aligns well with the results reported for o1 in Table 1. We denote the probability of correctly refusing a harmful query from StrongREJECT as $p_{\text{StrongREJECT}}^{(\text{o1})}$ and the probability of not overrefusing on XSTest as $p_{\text{XSTest}}^{(\text{o1})}$. To claim Pareto dominance, we need to show that $p_{\text{StrongREJECT}}^{(\text{o1})} > p_{\text{StrongREJECT}}^{(\text{Sonnet})}$ and $p_{\text{XSTest}}^{(\text{o1})} > p_{\text{XSTest}}^{(\text{Sonnet})}$. Thus, we perform two z -tests, one for each variable:

For StrongREJECT:

$$\begin{aligned}
 p_{\text{StrongREJECT}}^{(\text{o1})} &= \frac{274}{313} = 0.875, \\
 p_{\text{StrongREJECT}}^{(\text{Sonnet})} &= \frac{247}{313} = 0.789, \\
 p_{\text{StrongREJECT}} &= \frac{274 + 247}{313 + 313} = 0.832.
 \end{aligned}$$

Standard error:

$$\begin{aligned}
 SE &= \sqrt{p_{\text{StrongREJECT}}(1 - p_{\text{StrongREJECT}}) \left(\frac{1}{313} + \frac{1}{313} \right)} \\
 &= \sqrt{0.832 \times 0.168 \times \left(\frac{1}{313} + \frac{1}{313} \right)} \\
 &\approx 0.0297.
 \end{aligned}$$

z -statistic:

$$\begin{aligned}
 z &= \frac{p_{\text{StrongREJECT}}^{(\text{o1})} - p_{\text{StrongREJECT}}^{(\text{Sonnet})}}{SE} \\
 &= \frac{0.875 - 0.789}{0.0297} \\
 &\approx 2.90.
 \end{aligned}$$

This yields a p -value: ≈ 0.004 , which indicates a highly significant difference at $\alpha = 0.05$.

For overrefusal:

$$\begin{aligned} p_{\text{XSTest}}^{(\text{o1})} &= \frac{232}{250} = 0.928, \\ p_{\text{XSTest}}^{(\text{Sonnet})} &= \frac{222}{250} = 0.888, \\ p_{\text{XSTest}} &= \frac{232 + 222}{250 + 250} = 0.908. \end{aligned}$$

Standard error:

$$\begin{aligned} SE &= \sqrt{p_{\text{XSTest}}(1 - p_{\text{XSTest}}) \left(\frac{1}{250} + \frac{1}{250} \right)} \\ &= \sqrt{0.908 \times 0.092 \times \left(\frac{1}{250} + \frac{1}{250} \right)} \\ &\approx 0.0259. \end{aligned}$$

z -statistic:

$$\begin{aligned} z &= \frac{p_{\text{XSTest}}^{(\text{o1})} - p_{\text{XSTest}}^{(\text{Sonnet})}}{SE} \\ &= \frac{0.928 - 0.888}{0.0259} \\ &\approx 1.55. \end{aligned}$$

This yields a p -value: ≈ 0.12 , which is not significant at $\alpha = 0.05$.

Thus we cannot claim a Pareto improvement using experiments on these two datasets alone.

A.5.2. LIMITATIONS DUE TO AUTOMATED JUDGING.

The main body of the paper reports safety results on StrongREJECT using automated judges. However, the authors also include a human verification of jailbreaking performance in the appendix, which yields different results for StrongREJECT.

We perform an additional z -test to validate that o1 significantly improves jailbreaking robustness over Claude Sonnet 3.5 using the human-sourced data.

Given:

$$\begin{aligned} p_{\text{StrongREJECT}}^{(\text{o1})} &= 0.92, \\ p_{\text{StrongREJECT}}^{(\text{Sonnet})} &= 0.90, \\ p_{\text{StrongREJECT}} &= \frac{0.92 + 0.90}{1 + 1} = 0.91. \end{aligned}$$

Standard error:

$$\begin{aligned} SE &= \sqrt{p_{\text{StrongREJECT}}(1 - p_{\text{StrongREJECT}}) \left(\frac{1}{313} + \frac{1}{313} \right)} \\ &= \sqrt{0.91 \times 0.09 \times \left(\frac{1}{313} + \frac{1}{313} \right)} \\ &\approx 0.0229. \end{aligned}$$